Spontaneous Symmetry Breaking of Reference Frames as a Computational Cost Minimization Strategy

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Abstract

We investigate a conditional implementation model for bounded agents that process memory-like state under finite computational resources. Revised Paper I [1] supplies a conditional correlation-resource account; it does not establish universal memory necessity, a universal Markovian ceiling, or a Landauer charge for merely storing records.

The present paper therefore studies a declared raw D-channel archive processed under a finite throughput budget. Within that model, symmetric processing of all declared channels can exceed the available budget at a finite time, while a supplied coordinate selector can reduce the declared load. This is a conditional overload result for the stated archive and algorithm, not a theorem that every persistent agent must compress memory or break a symmetry.

Reference-frame selection is treated as a modelling strategy: a privileged basis or coordinate projection packages a lower-dimensional representation of the archive. The rate-distortion calculation gives a budget-dependent target dimension under the stated source, distortion, and budget assumptions. It does not prove that the "self" is a universal survival-optimal structure.

The projected representation is accompanied by four diagnostic labels: basis selection, frame drag, objective centering, and model incompleteness. Only basis selection follows from projection alone; the other diagnostics require additional controller, update, objective, or task assumptions. A simple drift model gives a visibility crossover for a fixed frame, motivating Paper III's calibration problem without establishing an undetectable universal Delusion Trap.

1 Introduction

1.1 Context: The Problem of Overload

Revised Paper I [1] gives an ELV-type accounting identity in which initial correlations can fund a later interval and proves a ceiling only for a fixed-Hamiltonian autonomous Davies baseline. It retracts both universal memory necessity and a Landauer charge for merely storing or maintaining records.

The present paper therefore begins with an independent implementation model: a raw, uncompressed D-channel archive processed under a finite throughput budget. Its overload is conditional on that archive and algorithm. The question is when a supplied coordinate selector can reduce this declared load, not whether every persistent agent must compress or break symmetry.

1.2 Position within the Series

This paper is the second of three constituting the **T-DOME** (Thermodynamic Dynamics of Observer-Memory Entanglement) framework, the third pillar of a three-paper program.

Framework	**Question**		**Result**	**Status**
HAFF [16, 17]	How does geometry emerge?	Ocean	Conditional algebra/dictionary models	Revised
Q-RAIF [19, 20]	What algebra must an observer have?	Fish	Clifford representation as model input	Revised
T-DOME I [1]	What can inter-epoch correlations fund?	Seed	Restricted baseline; accounting identity	Conditional
T-DOME II (this work)	When can a model select a frame?	Ego	Conditional coordinate selection	**This paper**
T-DOME III	When can residual statistics track drift?	Loop	Conditional Fisher calibration	Companion

The three T-DOME papers form a conditional modelling sequence rather than an irreversible theorem chain:

1. **Paper I (The Seed):** An ELV-type balance identifies initial correlations as one possible interval resource. Raw-archive overload is an implementation scenario, not a thermodynamic necessity.

2. **Paper II (The Ego, this work):** Unbounded memory under finite computational resources causes processing overload in the coordinate-processing model. This paper studies conditional feature selection; it does not prove universal SSB.

3. **Paper III (The Loop):** Uncorrected bias diverges from a changing environment. A self-referential calibration loop (monitoring one's own prediction error) resolves the bias but requires the system to "observe its own observation"—closing the self-calibration loop.

1.3 Relation to Q-RAIF

Revised Q-RAIF Papers A and B [19, 20] use a Clifford algebra $Cl(V, q)$ as a supplied representation choice; persistence does not select it. Paper C [21] proves only that the image

of a separately supplied homomorphism lies in the external algebra. This paper adopts both the Clifford coordinates and the representation map as model inputs.

The Clifford algebra $Cl(V, q)$, however, admits a non-trivial *automorphism group* $G = \mathrm{Aut}(Cl(V, q))$. In the absence of external constraints, all elements of G yield physically equivalent representations—the choice of basis within the algebra is a *gauge freedom*. This gauge freedom is the mathematical substrate of the symmetry considered in the conditional breaking analysis below.

The "ego" names a selected computational frame in that supplied representation. Whether a concrete optimum is a genuine gauge fixing is a model condition, not an inherited theorem.

1.4 Relation to HAFF Paper G

Revised HAFF Paper G [18] offers philosophy-of-science motivation and no fixed-point no-go theorem. It does not imply basis selection here. Any connection between its meta-level concern and this finite-budget model is motivation only.

1.5 Scope and Disclaimers

To prevent interpretational overreach, we state at the outset what this paper does *not* claim:

1. We do not claim that symmetry breaking is *sufficient* for persistence. Paper III addresses the additional requirements.

2. We do not claim that basis selection or spontaneous symmetry breaking is necessary for every bounded agent. The results concern a declared coordinate-retention model.

3. The term "ego" or "self" is used in the control-theoretic sense: a fixed reference frame within the agent's internal algebra. It carries no implication of consciousness or subjective experience.

4. A broader structural analogy with classical philosophical concepts of selfhood exists but is outside the scope of this paper.

Related work. The idea that bounded agents must compress their representations has roots in Simon's bounded rationality [8], Shannon's rate-distortion theory [6, 7], and Sims's rational inattention [10], which models finite-capacity decision-makers as solving a rate-distortion problem—precisely the economic counterpart of our $\mathcal{C}_{\mathrm{budget}}$ formalism. The information bottleneck [9] formalises relevance-weighted compression and has been applied to neural coding and deep learning. The role of decoherence in selecting preferred bases (pointer states) is well established via quantum Darwinism [14]; here it motivates one model-specific coordinate-selection rule. Measures of non-Markovianity and their thermodynamic consequences are reviewed in [15, 2]; the connection to survival was established in Paper I.

Summary of contributions. This paper separates three levels of result:

1. **Raw-channel load scaling law** a conditional accounting identity for a raw D-channel archive.

2. **Survival-weighted rate-distortion bound** a compactness-based existence result; the budget supplies only a capacity $k_{\max}$, not a universal optimal retained dimension.

3. **Delusion dynamics** a linear-drift visibility crossover plus an explicit two-mode example showing when gradient anti-correlation can occur.

Result-status ledger. The original Computational Ceiling and SSB-necessity chain is retracted ('TD-II-T3.5'). Raw-channel load is an FD/I accounting proposition; compression existence is conditional on a compact feasible model; SSB is a model interpretation. The Delusion Trap is downgraded ('TD-II-T6.7'): visibility crossover is exact in the declared two-mode geometry, while gradient sign and self-detection require extra dynamics.

2 Mathematical Preliminaries

2.1 Inherited Framework from Paper I

We briefly recall the key objects from Paper I [1] that the present work builds upon. The reader is referred to Paper I for full definitions and proofs.

Survival functional. For an open quantum system S coupled to an environment E at inverse temperature β, with dynamics Λ and external control protocol $H_{\mathrm{ctrl}}(t)$, the survival functional is

$$\mathcal{S}[\Lambda, \tau] := \Delta F - W[0, \tau], \tag{1}$$

where $\Delta F = F(\rho(\tau)) - F(\rho(0))$ is the change in non-equilibrium free energy and $W = \int_0^\tau \mathrm{tr}\Big(\rho(t)\, \dot{H}_{\mathrm{ctrl}}(t)\Big)\, dt$ is the work performed by the external protocol.

Restricted autonomous baseline. For a fixed-Hamiltonian Davies semigroup with no control, Paper I proves

$$\mathcal{S}[\Lambda^{\mathrm{M}}, \tau] \leq 0 \qquad \text{for all } \tau \geq 0. \tag{2}$$

It does not extend this result to driven Markovian agents.

Non-Markovian advantage identity. For arbitrary initial states:

$$\beta\, \mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}(\rho_E \| \rho_E^{\mathrm{th}}) - \beta\, \Delta\langle H_{\mathrm{ctrl}} \rangle - \beta\, \Delta\langle H_{\mathrm{int}} \rangle. \tag{3}$$

Conditional raw-archive model. Paper I retracts a universal Landauer maintenance bound. This paper independently posits an uncompressed record stream. Let h_μ be its *per-component* entropy rate [3], defined by

$$h_\mu := \lim_{T\to\infty} \frac{1}{T} H(X_{0:T}), \tag{4}$$

measuring the asymptotic information (in bits per unit time) generated by one declared record channel (we work in units where the sampling interval equals the environmental correlation time τ_E)[1] The predictive information I_{pred} is the *excess entropy* (excess entropy) [5, 4], defined as the mutual information between past and future of the environmental process:

$$I_{\text{pred}} := I\Big(\overleftarrow{X}; \overrightarrow{X}\Big) = H(\overrightarrow{X}) - H(\overrightarrow{X} \mid \overleftarrow{X}), \tag{5}$$

where $\overleftarrow{X}$ and $\overrightarrow{X}$ denote the semi-infinite past and future, respectively. For a stationary process, I_{pred} relates to h_μ via the entropy-rate decomposition $H(X_{1:T}) = I_{\text{pred}} + h_\mu T + o(1)$ as $T \to \infty$ when the relevant limits exist [3]. No lower bound $I_{\text{stored}} \geq \min(I_{\text{pred}}, h_\mu \tau)$ is inherited.

2.2 The Agent's Internal Algebra

As a modelling choice motivated by revised Q-RAIF [20, 21], the agent's internal control algebra is taken to be a Clifford algebra $\mathcal{O}_{\text{int}} = Cl(V, q)$ for a real vector space V equipped with a non-degenerate quadratic form q. The algebra satisfies the fundamental relation $v^2 = q(v)\,\mathbf{1}$ for all $v \in V$.

The *automorphism group*

$$G := \text{Aut}(Cl(V, q)) \tag{6}$$

is the group of algebra automorphisms that preserve the grading and quadratic form.[2] For $Cl(1, 3)$, G contains the spin group $\text{Spin}(1, 3) \cong SL(2, \mathbb{C})$ as a subgroup—a six-real-dimensional Lie group.

In the absence of computational constraints, all $g \in G$ yield physically equivalent descriptions of the agent's internal state. The choice of basis within $Cl(V, q)$ is a *gauge freedom*—the symmetry examined in the conditional coordinate-selection model below.

A representation map $\phi : Cl(V, q) \to \mathcal{A}_{\text{ext}}$ is supplied as an interface. Q-RAIF Paper C proves only the image-level algebraic statement once this map is given; it does not select a preferred $Cl(1, 3)$ target or preserve quadratic signature.

[1] For a continuous-valued process sampled at resolution b bits, h_μ includes the quantisation cost: $h_\mu = h_\mu^{(\text{diff})} + b\, f_s$, where $h_\mu^{(\text{diff})}$ is the differential entropy rate and f_s the sampling frequency. All raw-channel accounting identities in this paper use h_μ so defined.

[2] We use G as an effective symmetry group acting transitively on admissible frames. The detailed Lie-algebraic structure of G is not required for our results; only the existence of a non-trivial symmetry that may be broken under the additional encoder conditions in Remark 18. In concrete models, one may replace G by its image under the adjoint representation—typically $O(V, q)$ or a pin/spin subgroup.

Dimensional convention. Two distinct notions of dimension appear throughout:

Symbol	Meaning	Scaling
$n := \dim V$	number of generators (degrees of freedom)	—
$D := \dim Cl(V, q) = 2^n$	full multivector space (algebra basis size)	exponential in n

The raw-channel load model (Section 3) scales with D, not n; the distinction matters whenever one compares generator-level and algebra-level quantities.

2.3 Rate-Distortion Theory

We require the classical rate-distortion framework of Shannon [6].

Definition 1 (Rate-distortion function)**.** *Let X be a random source with distribution $p(x)$, $\hat{X}$ a reconstruction, and $d : \mathcal{X} \times \hat{\mathcal{X}} \to [0, \infty)$ a distortion measure. The* rate-distortion function *is*

$$R(D) := \min_{\substack{p(\hat{x}|x): \\ \mathbb{E}[d(X,\hat{X})] \leq D}} I(X; \hat{X}), \tag{7}$$

the minimum mutual information between source and reconstruction that achieves average distortion at most D.

$R(D)$ is a convex, non-increasing function of D with $R(0) = H(X)$ (lossless) and $R(D_{\max}) = 0$ (maximum distortion). It provides the fundamental limit on lossy compression [7]. The *information bottleneck* method of Tishby et al. [9] generalises this framework to the case where the relevant variable is not the source itself but a downstream prediction target—precisely the situation in our survival-weighted compression problem (Section 4.2). For source X, representation Z, and declared survival-relevant target Y, the bottleneck objective is

$$\min_{p(z|x)} I(X; Z) - \beta_{\text{IB}} I(Z; Y). \tag{8}$$

The coordinate-retention model below does not solve this unrestricted optimization. Rate-distortion and information bottleneck therefore provide comparison frameworks, not authority for the component count.

2.4 Bounded Rationality

Following Simon [8], we model computational limitations as a hard constraint on the agent's information processing rate.

Definition 2 (Computational budget)**.** *The agent's* computational budget $\mathcal{C}_{\text{budget}}$ *(measured in bits per unit time) is the maximum rate at which the agent can evaluate functions of its stored correlations. We assume $\mathcal{C}_{\text{budget}} < \infty$.*

Physically, finiteness of $\mathcal{C}_{\text{budget}}$ reflects the finite number of degrees of freedom in the agent's physical substrate: finite Hilbert space dimension, finite memory register size, and finite energy available for computation (Landauer's principle [12, 13]).

2.5 Fiber Bundle Formalism

The geometric setting for reference-frame selection is a principal fiber bundle.

Definition 3 (Gauge bundle)**.** *The* gauge bundle *is the principal G-bundle*

$$\pi : P \to M, \qquad G = \mathrm{Aut}(Cl(V, q)), \tag{9}$$

where:

- *M is the base space of* effective memory kernels*—equivalently, the space of induced sufficient-statistic processes accessible to the agent (a finite-dimensional manifold that admits local parametrisation by the environmental spectral-density couplings);*
- *G is the structure group acting transitively on admissible frames (see footnote 2 for the effective subgroup);*
- *the fiber $\pi^{-1}(\kappa)$ over a kernel $\kappa \in M$ is the G-orbit of equivalent algebraic representations (frames) for describing κ in $Cl(V, q)$;*
- *a* section *$\sigma : M \to P$ constitutes a global gauge-fixing policy—a systematic choice of reference frame for every kernel configuration.*

A *connection* on P specifies how the reference frame is parallel-transported as the agent's state evolves. The curvature of this connection measures the extent to which the reference frame "twists" along different paths through state space.

2.6 Standing Assumptions

Definition 4 (Standing Assumptions)**.** *Throughout this paper, the following conditions are assumed:*

(B1) ***Inherited framework.*** *All assumptions (A1)–(A5) of Paper I [1] remain in force (open quantum system coupled to a thermal bath, well-defined free energy, non-equilibrium initial state, finite-dimensional system Hilbert space, and weak-coupling or controlled-coupling regime). Additionally, the agent possesses an internal control algebra $\mathcal{O}_{\mathrm{int}} = Cl(V, q)$ with a supplied representation map $\phi : Cl(V, q) \to \mathcal{A}_{\mathrm{ext}}$ (Q-RAIF [21]). Neither the Clifford choice nor the map is derived from persistence.*

(B2) ***Finite computational budget.*** *The agent's information processing rate satisfies $\mathcal{C}_{\mathrm{budget}} < \infty$ (Definition 2).*

(B3) ***Raw-channel load model.*** *The supplied record process has D declared component channels, per-channel innovation rate $h_\mu > 0$, memory depth $\tau_{\mathrm{mem}} > 0$, and no shared-code or group-theoretic compression before the accounting in Section 3. This worst-case assumption, not environmental non-triviality alone, carries the load calculation.*

(B4) ***Survival imperative.*** *The agent's dynamics must maintain $\mathcal{S} \geq \mathcal{S}_{\min}$ over survival horizons $T \gg \tau_{\mathrm{mem}}$. This is a persistence constraint, not an optimization objective.*

(B5) ***Gauge symmetry of bare algebra.*** *The automorphism group $G = \mathrm{Aut}(Cl(V,q))$ is non-trivial ($G \neq \{e\}$). In the absence of computational constraints, all $g \in G$ yield physically equivalent descriptions.*

3 Raw-Channel Load Model

We now record the throughput of the raw-channel model in (B3). This is a bookkeeping identity for an explicitly compression-free representation, not a lower bound over all symmetric algorithms and not a consequence of Paper I.

3.1 The Information Processing Inequality for Bounded Agents

Accounting convention. To ensure dimensional consistency throughout, we distinguish two quantities:

- $\mathcal{C}_{\mathrm{budget}}$: the agent's processing *rate* (bits per unit time).
- $\mathcal{I}_{\mathrm{proc}}(\tau)$: the total information (bits) that must be processed per evaluation cycle when the memory archive has depth τ.

The agent must complete one evaluation cycle per environmental correlation time τ_E. The *processing rate* required for a memory depth τ is

$$\mathcal{R}_{\mathrm{proc}}(\tau) := \frac{\mathcal{I}_{\mathrm{proc}}(\tau)}{\tau_E}. \tag{10}$$

Paralysis occurs when $\mathcal{R}_{\mathrm{proc}}(\tau_{\mathrm{mem}}) > \mathcal{C}_{\mathrm{budget}}$. Hereafter we measure time in units of τ_E (i.e., set $\tau_E = 1$), so that rates and per-cycle information quantities are numerically equal.

Definition 5 (Symmetric processing)**.** *An agent processes its memory* symmetrically *if both its cost functional $\mathcal{C}[\cdot]$ and its distortion measure $D(\cdot)$ are G-invariant: $\mathcal{C}[g \cdot \mathcal{K}] = \mathcal{C}[\mathcal{K}]$ and $D(g \cdot \mathcal{F}) = D(\mathcal{F})$ for every $g \in G = \mathrm{Aut}(Cl(V,q))$. In operational terms: for every stored correlation c_i in the memory kernel $\mathcal{K}(t,s)$ and every $g \in G$, the cost of evaluating c_i equals the cost of evaluating $g \cdot c_i$, and no basis direction is* a priori *preferred for survival evaluation.*

Remark 6 (Operational meaning of processing rate)**.** *We define the processing rate $\mathcal{R}_{\mathrm{proc}}$ as an* information-throughput *measure: the number of algebraic components that must be updated per unit time, multiplied by the innovation rate h_μ per component. It captures the* bandwidth *cost of maintaining an internal representation, not the algorithmic gate complexity of individual operations.*

Proposition 7 (Raw-channel load accounting)**.** *Under (B3), let $n = \dim V$ and $D = \dim Cl(V,q) = 2^n$. A raw archive of depth τ_{mem} contains, by the model definition,*

$$\mathcal{I}_{\mathrm{archive}} = D\, h_\mu\, \tau_{\mathrm{mem}} \qquad \textit{bits.} \tag{11}$$

Its incremental input rate is Dh_μ bits per unit time. If the declared algorithm rescans the entire archive once per environmental correlation time τ_E, its required throughput is

$$\mathcal{R}_{\text{sweep}} = \frac{Dh_\mu \tau_{\text{mem}}}{\tau_E}. \tag{12}$$

For a processing budget $\mathcal{C}_{\text{budget}}$, this particular full-sweep algorithm fits only when

$$\tau_{\text{mem}} \leq \tau_{\max} := \frac{\mathcal{C}_{\text{budget}} \tau_E}{Dh_\mu}. \tag{13}$$

Proof. All three statements follow by multiplying the declared number of channels by their per-channel innovation rate and then converting stored bits per sweep to bits per unit time. No lower bound over compressed, streaming, sufficient-statistic, or group-equivariant algorithms is claimed. □

Remark 8 (Conditional component-count ratio)**.** *Within the same raw-coordinate implementation, comparing a D-channel sweep with a k-channel sweep gives the component-count ratio D/k. This is not a tax imposed by gauge invariance on all algorithms.*

Remark 9 (Effective vs. full dimension)**.** *The ceiling uses $D = \dim Cl(V, q) = 2^n$, the full multivector dimension. In practice, the environment may couple to only a subset of grades (e.g., grade-1 generators), yielding an effective dimension $D_{\text{eff}} \leq D$. For a* structured *environment where the agent knows which grades are active, the ceiling can be tightened to $\mathcal{R}_{\text{proc}} \gtrsim h_\mu \cdot D_{\text{eff}}$. The raw-channel condition in (B3) represents this worst case. A structured environment or algorithm changes the model rather than falling under a universal ceiling.*

3.2 Processing Collapse

Remark 10 (Raw full-sweep overload)**.** *If $\tau_{\text{mem}} > \tau_{\max}$, the full-sweep algorithm in Proposition 7 misses its declared cycle deadline. This does not imply that every record is useless or that another streaming/compressed algorithm fails.*

Remark 11 (Comparison with revised Paper I)**.** *Paper I now treats storage/reset cost as implementation-dependent. The present overload is likewise conditional on a raw full-sweep algorithm. Compression is one response; neither paper proves it must take the form of symmetry breaking.*

4 Conditional Frame-Selection Model

4.1 Reference Frame as Gauge Fixing

Definition 12 (Reference frame)**.** *A* reference frame *$\mathcal{F}$ is a section $\sigma : M \to P$ of the gauge bundle (Definition 3). Choosing σ is equivalent to selecting a preferred orthonormal basis $\{e_1, \ldots, e_n\}$ of the generating vector space V at each point in state space M, thereby fixing the gauge freedom of $Cl(V, q)$.*

Definition 13 (Projected memory kernel)**.** *Given a reference frame $\mathcal{F}$, let $V_{\mathrm{fg}}(\mathcal{F}) \subset Cl(V,q)$ be a k-dimensional* foreground subspace *selected by the declared coordinate model, with $0 \le k \le D$. Let $\Pi_{\mathcal{F}}$ denote the orthogonal projection onto $V_{\mathrm{fg}}(\mathcal{F})$ with respect to the trace inner product $\langle A, B\rangle := \mathrm{tr}\big(A^\dagger B\big)$. The* projected memory kernel *is*

$$\mathcal{K}_{\mathcal{F}}(t,s) := \Pi_{\mathcal{F}}\, \mathcal{K}(t,s)\, \Pi_{\mathcal{F}}. \tag{14}$$

The complementary projection $\Pi_{\mathcal{F}}^{\perp} = \mathbf{1} - \Pi_{\mathcal{F}}$ defines the background subspace $V_{\mathrm{bg}}(\mathcal{F})$*. The decomposition $Cl(V,q) = V_{\mathrm{fg}} \oplus V_{\mathrm{bg}}$ is determined by $\mathcal{F}$, not by any* a priori *ordering of basis vectors.*

4.2 A Coordinate-Compression Model

We now define a restricted coordinate-compression problem and compare it with rate-distortion theory.

Processing rate of a frame. If the agent retains k algebraic components (the foreground subspace V_{fg}), each generating h_μ bits per unit time, the processing rate of frame $\mathcal{F}$ is

$$R_{\mathcal{F}}(k) \;=\; k \cdot h_\mu \qquad \text{(bits per unit time)}. \tag{15}$$

The budget constraint $R_{\mathcal{F}} \le \mathcal{C}_{\mathrm{budget}}$ thus bounds the number of maintainable components. Define the capacity

$$k_{\max} := \min\left(D, \left\lfloor \frac{\mathcal{C}_{\mathrm{budget}}}{h_\mu} \right\rfloor\right). \tag{16}$$

This is a maximum feasible component count, not yet an optimizer.

Definition 14 (Survival distortion)**.** *The* survival distortion *of a reference frame $\mathcal{F}$ is*

$$D(\mathcal{F}) := \mathbb{E}_\xi\left[\ell\big(\mathcal{S}_{\mathrm{full}}(\xi) - \mathcal{S}_{\mathcal{F}}(\xi)\big)\right], \tag{17}$$

where ξ denotes environmental realizations, $\ell : \mathbb{R} \to [0,\infty)$ is a convex, non-decreasing loss function (we use squared error $\ell(x) = x^2$ throughout), $\mathcal{S}_{\mathrm{full}}(\xi)$ is the survival functional evaluated using the full memory kernel $\mathcal{K}(t,s)$, and $\mathcal{S}_{\mathcal{F}}(\xi)$ is evaluated using the projected kernel $\mathcal{K}_{\mathcal{F}}(t,s)$.

Remark 15 (Information-theoretic objects)**.** *Strictly speaking, rate-distortion theory and mutual information apply to stochastic processes, not to superoperator kernels directly. Throughout Sections 4–7, $I(\mathcal{K}_{\mathcal{F}};\, \mathcal{K})$ is shorthand for $I(\hat{X};\, X)$, where $X = \{c_i(t)\}_{i=1}^{D}$ is the sufficient-statistic record process induced by the full kernel $\mathcal{K}$ acting on the agent's internal coordinates, and $\hat{X} = \{c_i(t)\}_{i \in V_{\mathrm{fg}}}$ is the projected record induced by $\mathcal{K}_{\mathcal{F}}$. The distortion measure* (17) *acts on the survival functional $\mathcal{S}$ evaluated on these records.*

Proposition 16 (Existence in the coordinate model)**.** *For each $0 \le k \le k_{\max}$, let $\mathfrak{F}_k$ be a compact set of admissible k-component frames. Assume the feasible union $\mathfrak{F}_{\le k_{\max}}$ is nonempty and the distortion $D(\mathcal{F})$ is lower semicontinuous. Then*

$$\mathfrak{F}^* := \operatorname*{arg\,min}_{\mathcal{F} \in \mathfrak{F}_{\le k_{\max}}} D(\mathcal{F}) \tag{18}$$

is nonempty. Every optimizer has some retained dimension $k_{\text{opt}} \leq k_{\max}$. It follows only by counting that its discarded component fraction is

$$1 - \frac{k_{\text{opt}}}{D}. \tag{19}$$

Budget saturation $k_{\text{opt}} = k_{\max}$ requires the additional strict-improvement condition that every feasible added component lowers the best attainable distortion. Without it, unused capacity is possible.

Proof. The feasible union is compact (a finite union of compact sets), so a lower-semicontinuous function attains its minimum. The dimension and discard fraction statements are definitions. Mere non-increase of distortion does not imply strict improvement or budget saturation. □

Remark 17 (Relation to Shannon and information bottleneck)**.** *Proposition 16 is not a Shannon converse or an information-bottleneck solution. Those theories optimize over stochastic encoders; this proposition optimizes over a supplied compact family of coordinate projections. The original theorem claiming $k^* = \lfloor \mathcal{C}_{\text{budget}}/h_\mu \rfloor$ as the universal optimum is withdrawn.*

For the remainder of the paper, k^* denotes the retained dimension of one chosen optimizer from $\mathfrak{F}^*$, when such a choice is made. Thus $k^* = k_{\text{opt}} \leq k_{\max}$; it is not identified with $\lfloor \mathcal{C}_{\text{budget}}/h_\mu \rfloor$ unless the stated strict-improvement condition is separately verified.

4.3 Conditional Symmetry-Breaking Interpretation

Remark 18 (Conditional symmetry-breaking interpretation)**.** *Suppose, in addition to Proposition 16, that (i) every admissible compressor is a proper coordinate projection, (ii) the effective group G acts on the candidate frames, and (iii) no G-invariant feasible encoder attains the required distortion. Then choosing any feasible optimizer privileges a foreground subspace of the D-dimensional algebra representation and breaks that model symmetry operationally. This conclusion is conditional on the encoder class; it is not a universal necessity theorem.*

If $D(\mathcal{F}) \neq D(\mathcal{F}')$ for distinct frames, that condition supports uniqueness of a global minimizer, not multiple local minima. Multiple "ego attractors" require a separate non-convex landscape with several demonstrated minima. The original SSB theorem and its generic non-uniqueness claim are withdrawn.

4.4 The Four Bias Terms

Definition 19 (Four frame-dependent diagnostics)**.** *For a declared coordinate projection $\Pi_{\mathcal{F}}$ and a specified controller, we use the following four labels as diagnostics. They organize possible departures from a symmetric representation; they are not asserted to be universal consequences of gauge fixing.*

(i) ***Basis selection bias*** *($\mathcal{B}_{\text{select}}$): The choice of $\{e_1, \ldots, e_{k^*}\}$ privileges certain algebraic components over others. Information aligned with the chosen basis is processed efficiently; misaligned information is discarded or distorted.* Observable consequence: *systematic blindness to off-basis environmental perturbations (orthogonal masking).*

(ii) ***Frame drag*** *($\mathcal{B}_{\text{frame}}$): If the update rule transports the current frame with memory or a restoring penalty, it may prefer states near that choice.* Possible observable consequence: *hysteresis in belief updating and lag behind environmental shifts. Holonomy alone does not supply a restoring force.*

(iii) ***Objective centering*** *($\mathcal{B}_{\text{center}}$): A projected surrogate objective depends only on the retained coordinates. Unless it is sufficient for the full objective, its stationary points need not be global stationary points.* Possible observable consequence: *failure to detect optima whose relevant directions lie in the background subspace.*

(iv) ***Model incompleteness*** *($\mathcal{B}_{\text{inc}}$): The compression from $Cl(V, q)$ to V_{fg} is lossy. The discarded components V_{bg} may contain correlations absent from the retained representation.* Possible observable consequence: *underestimation of total uncertainty when those correlations matter to the task.*

Only the first item follows from the projection alone: components in $\ker \Pi_{\mathcal{F}}$ are removed. Frame drag requires a stateful update law, objective centering requires a specified surrogate loss, and overconfidence requires task-relevant information in the discarded coordinates. Proposition 16 supplies no generic information fraction for these diagnostics.

Bias	**Origin**	**Observable consequence**	**Determines**
$\mathcal{B}_{\text{select}}$ (selection)	projection $\Pi_{\mathcal{F}}$	Systematic blindness to off-basis perturbations (orthogonal masking)	*what* is seen
$\mathcal{B}_{\text{frame}}$ (frame drag)	bundle connection / holonomy	Hysteresis in belief updating; model lags behind rapid drift	*duration*
$\mathcal{B}_{\text{center}}$ (centering)	$\nabla \mathcal{S} \in V_{\text{fg}}$	Local frame-relative optima; global background optima invisible	*target*
$\mathcal{B}_{\text{inc}}$ (incompleteness)	lossy compression $k^* \ll D$	Underestimation of thermodynamic uncertainty (structural overconfidence)	*blind spot*

Table 1: The four bias terms of the broken phase. Only the first follows from projection alone; the others require the additional update, objective, or task assumptions stated in the text.

Remark 20 (Nature of the bias terms)**.** *The four bias terms (Table 1) are not pathologies. They are diagnostic names for distinct ways in which a projected controller may depart from a symmetric representation. Projection alone gives basis selection; the remaining terms become live only when the corresponding update, objective, or task structure is specified.*

5 Emergent Structure: The Architecture of Ego

We consolidate the gauge-fixed compressed representation into a single mathematical object. Throughout this section, "ego" is used purely as shorthand for a gauge-fixed compressed representation; no claims about phenomenal consciousness, subjective experience, or qualia are intended or implied.

Definition 21 (Ego)**.** *The* ego *of an agent satisfying* (B1)–(B5) *is the pair*

$$\mathfrak{E} := \big(\mathcal{F}^*,\ V_{\mathrm{fg}}^*\big), \tag{20}$$

where $\mathcal{F}^* \in \mathfrak{F}^*$ (Proposition 16) is a chosen frame (providing the coordinate system) and $V_{\mathrm{fg}}^* := V_{\mathrm{fg}}(\mathcal{F}^*)$ is the k^*-dimensional foreground subspace selected within the coordinate model (providing the compression). The projected memory kernel $\mathcal{K}_{\mathfrak{E}} := \Pi_{V_{\mathrm{fg}}^*}\,\mathcal{K}\,\Pi_{V_{\mathrm{fg}}^*}$ is induced by this pair. All bias terms, distortion bounds, and delusion dynamics are functions of $\mathfrak{E}$.

5.1 The Ego as a Fiber Bundle Section

The reference frame $\mathcal{F}$, understood as a section $\sigma : M \to P$, is the mathematical object we call the *ego*. It has three key properties:

Smoothness. The section σ varies continuously with the agent's state $\rho \in M$. Small changes in ρ produce small changes in the preferred basis—the ego is not a discrete switch but a smooth deformation of perspective.

Holonomy. If the agent's state traces a closed loop $\gamma : [0, 1] \to M$ with $\gamma(0) = \gamma(1) = \rho_0$, the parallel-transported frame need not return to its initial value:

$$\sigma(\gamma(1)) = \mathrm{Hol}(\gamma) \cdot \sigma(\gamma(0)), \tag{21}$$

where $\mathrm{Hol}(\gamma) \in G$ is the holonomy of the connection around γ. Non-trivial holonomy means the agent can "learn"—its reference frame shifts after a complete cycle of experience.

Topological obstruction. In general, a *global* section $\sigma : M \to P$ may not exist. The obstruction is measured by the characteristic classes of the bundle P. When a global section does not exist, the ego must have "singularities"—states where the preferred basis is undefined or discontinuous. This connects to the crisis of Paper III: the delusion trap can be understood as the agent approaching a topological obstruction of its own reference frame.

5.2 The Effective Survival Functional

Proposition 22 (Survival decomposition)**.** *In the broken phase, the survival functional decomposes as*

$$\mathcal{S} = \mathcal{S}_{\text{vis}}(\mathcal{F}) + \mathcal{S}_{\text{hid}}(\mathcal{F}), \tag{22}$$

where:

- $\mathcal{S}_{\text{vis}}(\mathcal{F})$ *is the contribution from the foreground subspace* V_{fg}*, computable within the agent's reference frame;*
- $\mathcal{S}_{\text{hid}}(\mathcal{F})$ *is the contribution from the background subspace* V_{bg}*, invisible to the agent.*

The agent maximizes $\mathcal{S}_{\text{vis}}$ *while being structurally blind to* $\mathcal{S}_{\text{hid}}$*.*

Proof. The survival functional $\mathcal{S} = \Delta F - W$ depends on $\rho(t)$, which is a function of the full memory kernel $\mathcal{K}(t,s)$. Decomposing $\mathcal{K} = \Pi_{\mathcal{F}}\mathcal{K}\,\Pi_{\mathcal{F}} + \Pi_{\mathcal{F}}^{\perp}\mathcal{K}\,\Pi_{\mathcal{F}}^{\perp} + \text{cross terms}$, the leading contributions are $\mathcal{S}_{\text{vis}} := \mathcal{S}[\Pi_{\mathcal{F}}\mathcal{K}\,\Pi_{\mathcal{F}}]$ and $\mathcal{S}_{\text{hid}} := \mathcal{S} - \mathcal{S}_{\text{vis}}$ (collecting background and cross terms). The agent computes only $\mathcal{S}_{\text{vis}}$, as the projected kernel $\mathcal{K}_{\mathcal{F}}$ discards all background components. □

5.3 The Computational Speedup

Proposition 23 (Ego dividend)**.** *After symmetry breaking, the computational cost of processing memory drops from* $\mathcal{C}_{\text{proc}} \sim h_\mu \cdot \tau_{\text{mem}} \cdot D$ *(symmetric case,* $D = \dim Cl(V,q)$*) to*

$$\mathcal{C}_{\text{proc}}^{(\mathcal{F})} \sim h_\mu \cdot \tau_{\text{mem}} \cdot k^*. \tag{23}$$

The speedup factor is

$$\frac{D}{k^*} = \frac{2^n}{k^*}. \tag{24}$$

This is the computational advantage of reference-frame selection. For $Cl(1,3)$ ($D = 16$) with $k^* = 2$, the speedup is $8\times$. For higher-dimensional algebras, the speedup grows exponentially in n.

5.4 Uniform-Channel Information Trade-off

Theorem 24 (Uniform-channel predictive-loss relation)**.** *Let* $X = \{c_i(t)\}_{i=1}^{D}$ *denote the full stochastic record process induced by the memory kernel* $\mathcal{K}$ *on the agent's internal coordinates, and let* $\hat{X} = \{c_i(t)\}_{i\in V_{\text{fg}}}$ *denote the projected record retained by the ego. The mutual information between compressed and full records satisfies*

$$I(\hat{X};\, X) \;\le\; H(\hat{X}) \;\le\; k^* \cdot h_\mu \cdot \tau_{\text{mem}}. \tag{25}$$

Under the additional assumption that I_{pred} (5) *is approximately uniformly distributed across the* D *algebraic components in the symmetric phase*[3]*, define the predictive information lost*

[3]This "uniformity assumption" is the information-theoretic counterpart of the raw-channel condition (B3). When some components carry disproportionately more predictive information, the bound tightens or loosens depending on the alignment between V_{fg} and the high-information subspace.

by projection as

$$I_{\text{lost}}^{\text{pred}} := I(X_{\text{past}}; X_{\text{future}}) - I(\hat{X}_{\text{past}}; X_{\text{future}}) \approx \left(1 - \frac{k^*}{D}\right) I_{\text{pred}}, \tag{26}$$

up to the error in the uniform-allocation approximation. This is a model relation, not a distribution-free lower bound.

Proof. By the data processing inequality, $I(\hat{X}; X) \leq H(\hat{X})$. The projected record $\hat{X}$ has k^* components, each carrying at most h_μ bits per unit time over a window of τ_{mem}, giving $H(\hat{X}) \leq k^* \cdot h_\mu \cdot \tau_{\text{mem}}$ [7]. This yields (25). Data processing also gives $I(\hat{X}_{\text{past}}; X_{\text{future}}) \leq I(X_{\text{past}}; X_{\text{future}}) = I_{\text{pred}}$. Under the uniformity assumption, each of the D components carries $\sim I_{\text{pred}}/D$, so the k^* retained components account for $\sim (k^*/D)\, I_{\text{pred}}$. Equation (26) follows by subtraction under that allocation assumption. □

Remark 25 (The price of selfhood)**.** *Equation* (26) *quantifies predictive information loss for the uniform raw-channel model. It does not say that every k^*-coordinate encoder loses this fraction, or that the selected encoder is globally optimal over all stochastic encoders.*

6 Worked Example: Qubit in a Two-Channel Bath

6.1 Model Setup

We extend Paper I's spin-boson model to demonstrate symmetry breaking explicitly. Consider a qubit ($\dim \mathcal{H}_S = 2$) with internal algebra $Cl(0,2) \cong \mathbb{H}$ (the quaternions, $\dim = 4$).

Symbol mapping. The general framework of Sections 3–5 specialises as follows:

General	This example	Value
$Cl(V,q)$	$Cl(0,2) \cong \mathbb{H}$	$D = 4$
$G = \text{Aut}(Cl(V,q))$	$SO(3)$	acting on $\{\mathbf{i},\mathbf{j},\mathbf{k}\}$
$\mathcal{C}_{\text{budget}}$	$2\,h_\mu$	bits/time
$k_{\max}$ (capacity)	$\lfloor 2h_\mu/h_\mu \rfloor = 2$	components
V_{fg}	span$\{1,\mathbf{k}\}$	dephasing subspace
V_{bg}	span$\{\mathbf{i},\mathbf{j}\}$	dissipative subspace
$\tau_{\max}$ (Prop. 7)	$2h_\mu/(4h_\mu) = 0.5$	ω_0^{-1}

The qubit is coupled to a bosonic environment through *two* independent channels:

- A *dephasing channel* via σ_z, with spectral density

$$J_z(\omega) = \frac{2\lambda_z\, \gamma_z\, \omega}{\omega^2 + \gamma_z^2} \qquad \text{(Lorentz–Drude)}, \tag{27}$$

  producing a memory kernel $\mathcal{K}_z(t,s)$ with non-Markovian backflow.

Quantity	Symbol	Value	Theorem check
Full dimension	D	4	model input
Entropy rate	h_μ	1.0 (normalised)	per-component rate
Budget	$\mathcal{C}_{\text{budget}}$	$2\,h_\mu$	Def. 2
Raw update check	$h_\mu D$ vs $\mathcal{C}_{\text{budget}}$	$4 > 2$	full raw update does not fit
Capacity	$k_{\max}$	$\lfloor 2/1 \rfloor = 2$	budget identity
Discard fraction	$1 - k^*/D$	$1/2 = 50\%$	Thm. 24
Full-sweep depth	$\tau_{\max}$	$2/(4) = 0.5$	Eq. (13)

Table 2: Parameter mapping for the two-channel qubit example. This example chooses a two-component frame and verifies that it fits the budget; it does not prove symmetry-breaking necessity or budget saturation for every optimum.

- A *dissipative channel* via σ_x, with spectral density

$$J_x(\omega) = \frac{2\lambda_x\,\gamma_x\,\omega}{\omega^2 + \gamma_x^2} \qquad \text{(Lorentz–Drude)}, \tag{28}$$

  producing a memory kernel $\mathcal{K}_x(t,s)$.

The full memory kernel is $\mathcal{K}(t,s) = \mathcal{K}_z(t,s) \oplus \mathcal{K}_x(t,s)$, and the quaternionic algebra $\mathbb{H} = \text{span}\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ has automorphism group $G = \text{Aut}(\mathbb{H}) \cong SO(3)$ (rotations of the pure quaternion subspace).

Parameters. We set $\omega_0 = 1$ (energy unit), $\lambda_z = 1$, $\gamma_z = 0.5$ (underdamped, strong non-Markovian effects in the dephasing channel), $\lambda_x = 0.3$, $\gamma_x = 5.0$ (overdamped, approximately Markovian in the dissipative channel), and the low-temperature regime $\beta\omega_0 \gg 1$.

Computational budget. The agent has $\mathcal{C}_{\text{budget}} = 2\,h_\mu$ bits per unit time—sufficient to track two components of $\mathbb{H}$ but not all four.

Parameter-to-model mapping. Table 2 collects the example parameters and checks the raw full-sweep budget.

6.2 Raw Full-Sweep Budget Limit

In the declared raw full-sweep implementation, the agent tracks all four quaternionic components $\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ simultaneously. The computational cost is

$$\mathcal{C}_{\text{proc}} = h_\mu \cdot \tau_{\text{mem}} \cdot D = 4\,h_\mu \cdot \tau_{\text{mem}}, \qquad D := \dim Cl(0,2) = 4. \tag{29}$$

The maximum raw full-sweep archive depth is

$$\tau_{\text{par}} = \frac{\mathcal{C}_{\text{budget}}}{h_\mu \cdot D} = \frac{2\,h_\mu}{4\,h_\mu} = 0.5 \qquad \text{(in units of } \omega_0^{-1}\text{)}. \tag{30}$$

Beyond $\tau_{\text{mem}} = 0.5\,\omega_0^{-1}$, this uncompressed implementation does not fit the stated budget. This does not exclude other symmetric compressed algorithms.

6.3 Frame Selection: Choosing σ_z

For comparison, we choose a frame that selects σ_z as the privileged basis direction, retaining the $\{1, \mathbf{k}\}$ subspace (the dephasing channel) as foreground and discarding $\{\mathbf{i}, \mathbf{j}\}$ (the dissipative channel) as background:

$$\mathbb{H} = \underbrace{\text{span}\{1, \mathbf{k}\}}_{V_{\text{fg}}\ (k^*=2)} \oplus \underbrace{\text{span}\{\mathbf{i}, \mathbf{j}\}}_{V_{\text{bg}}}. \tag{31}$$

Why σ_z? The dephasing channel ($\lambda_z = 1$, $\gamma_z = 0.5$) is strongly non-Markovian and carries the dominant survival-relevant information (the backflow revivals that enable $\mathcal{S} > 0$, as demonstrated in Paper I). The dissipative channel ($\lambda_x = 0.3$, $\gamma_x = 5.0$) is approximately Markovian and contributes primarily to decoherence—its survival value is negative.

This choice coincides with the *pointer basis* selected by environmental decoherence (quantum Darwinism [14]): the σ_z eigenstates are the states that survive decoherence and become redundantly encoded in the environment. The ego "accepts the suggestion" of decoherence, aligning its computational resources with the environmentally stable basis.

6.4 The Broken Phase: Effective Processing

In the broken phase, the projected memory kernel $\mathcal{K}_{\mathcal{F}} = \mathcal{K}_z$ retains only the dephasing-channel dynamics. The computational cost drops to

$$\mathcal{C}_{\text{proc}}^{(\mathcal{F})} = h_\mu \cdot \tau_{\text{mem}} \cdot k^* = 2\, h_\mu \cdot \tau_{\text{mem}}, \tag{32}$$

exactly half the symmetric cost (29). This implementation can process a raw archive up to depth $\tau_{\text{mem}} = 1\,\omega_0^{-1}$ before reaching its budget—twice the full-sweep depth of the four-channel baseline.

The survival functional in the broken phase is

$$\mathcal{S}_{\text{vis}}(\mathcal{F}) = \mathcal{S}[\mathcal{K}_z], \tag{33}$$

which gives the plotted positive backflow proxy at the first revival. This proxy comparison is not a proof of thermodynamic optimality.

The hidden component $\mathcal{S}_{\text{hid}} = \mathcal{S}[\mathcal{K}_x]$ is the survival contribution from the dissipative channel, which the agent can no longer evaluate. For the chosen parameters, $|\mathcal{S}_{\text{hid}}| \ll |\mathcal{S}_{\text{vis}}|$ (the dissipative channel contributes primarily negative survival value), so the distortion is small.

6.5 Quantitative Evaluation

We now evaluate the declared implementation comparison explicitly. Each channel's decoherence function follows from the exact $T \to 0$ solution of the Lorentz–Drude pure-dephasing model [2, 1]:

$$p_\alpha(t) = e^{-\gamma_\alpha t/2}\left[\cos(\Omega_\alpha t) + \frac{\gamma_\alpha}{2\Omega_\alpha}\,\sin(\Omega_\alpha t)\right], \quad \Omega_\alpha := \tfrac{1}{2}\sqrt{4\lambda_\alpha\gamma_\alpha - \gamma_\alpha^2}, \tag{34}$$

for $\alpha \in \{z, x\}$. When $4\lambda_\alpha\gamma_\alpha < \gamma_\alpha^2$ (the overdamped regime), Ω_α becomes imaginary and the trigonometric functions are replaced by hyperbolic functions (monotonic decay, no backflow).

For our parameters:

- **z-channel** ($\lambda_z = 1$, $\gamma_z = 0.5$): $\Omega_z = \frac{1}{2}\sqrt{1.75} \approx 0.661$. Underdamped; $|p_z(t)|$ exhibits oscillatory backflow.
- **x-channel** ($\lambda_x = 0.3$, $\gamma_x = 5.0$): Discriminant $4\lambda_x\gamma_x - \gamma_x^2 = 6 - 25 = -19 < 0$. Overdamped; $|p_x(t)|$ decays monotonically with no backflow.

For this model we plot the proxy used in Paper I, $\beta\,\mathcal{S} \propto |p(t)|^2 - 1$ (for the pure-dephasing model with maximally coherent initial state and pointer-basis measurement), evaluated for each channel independently. Backflow intervals are those where $d|p_\alpha|/dt > 0$. The proxy increases there; interpreting that increase as thermodynamic survival requires the initial-state and interaction-energy conditions of the revised Paper I accounting identity.

Model comparison. For the z-channel with $\gamma_z = 0.5$, the first backflow interval begins at $t^* \approx 2.9\,\omega_0^{-1}$—well after the raw full-sweep depth $\tau_{\max} = 0.5\,\omega_0^{-1}$. The stipulated raw full-sweep implementation stops at its budget limit before the plotted backflow interval. The selected z-channel implementation remains within its declared component budget and records the three plotted revivals.

The cumulative backflow harvested by the ego agent (Figure 1(b)) totals approximately 0.10 (in dimensionless $\beta\,\mathcal{S}$ units) over $t \in [0, 15\,\omega_0^{-1}]$. The raw baseline records zero after its stipulated stopping point. The result compares these two implementations only; it is not an infinite performance ratio against all symmetric algorithms and does not show that backflow is accessible only after symmetry breaking.

Accounting check. The raw update load is $h_\mu D = 4h_\mu$ while the budget is $2h_\mu$, so that declared representation does not fit. The example then *chooses* the two-component dephasing frame ($k^* = 2$); for that choice, $R_{\mathcal{F}} = 2\,h_\mu = \mathcal{C}_{\text{budget}}$. Saturation is an example property, not a general consequence of monotonic distortion.

6.6 The Pointer-State Connection

In this example the chosen basis coincides with an einselection (environment-induced superselection) basis of the dephasing model [14]. The coordinate optimization can select pointer-aligned components when the declared distortion assigns them the highest survival value per bit. This alignment is verified only within a concrete model and is not generic rate-distortion theorem.

Remark 26 (Decoherence as symmetry-breaking catalyst)**.** *The environment does not* force *a specific gauge fixing. A pointer-aligned frame is selected only if the declared distortion and encoder class rank it above the alternatives. The present example chooses that ranking; it does not derive it from bounded computation.*

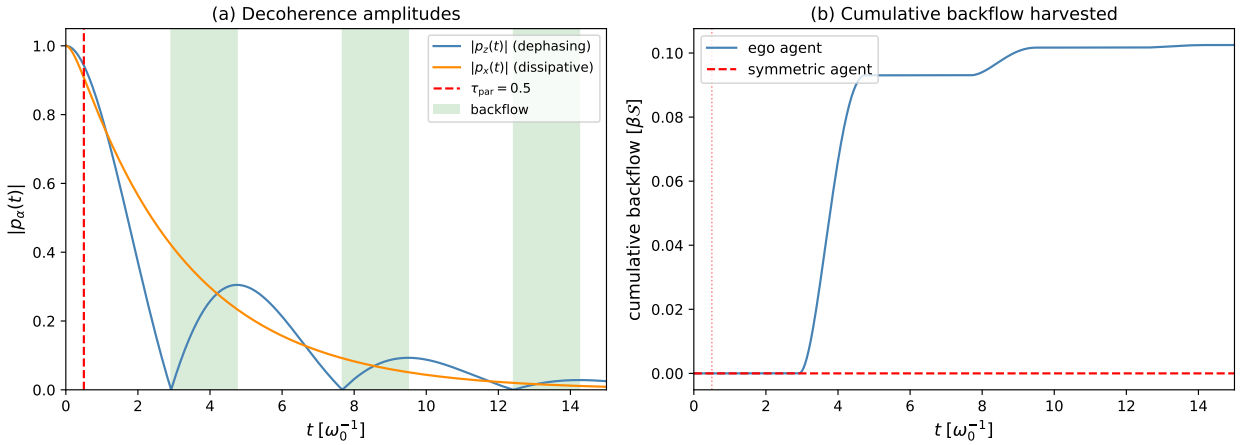


Figure 1: Two-channel qubit model (Section 6) with Lorentz–Drude spectral density. **Parameters:** $\omega_0 = 1$ (energy unit); $\lambda_z = 1$, $\gamma_z = 0.5$ (dephasing, non-Markovian); $\lambda_x = 0.3$, $\gamma_x = 5.0$ (dissipative, $\sim$Markovian); $\mathcal{C}_{\text{budget}} = 2\,h_\mu$. **Units:** time in ω_0^{-1}. **Regime:** low temperature ($\beta\omega_0 \gg 1$); using the standard $T \to 0$ analytic expression (34) [2]. **(a)** Decoherence amplitudes $|p_z(t)|$ (blue, non-Markovian, with backflow in green bands) and $|p_x(t)|$ (orange, monotonic decay). Red dashed: raw full-sweep depth $\tau_{\max} = 0.5$. **(b)** Cumulative backflow harvested. Blue: ego agent (broken $\to \sigma_z$) exploits all three revival intervals. Red dashed: the stipulated raw full-sweep agent stops at $\tau_{\max}$; all plotted backflow occurs later. The growing gap is the *ego dividend*.

7 The Cost of Ego

Coordinate selection reduces the declared raw load but can introduce a blind subspace. A fixed reference frame is a *static* gauge choice in a *dynamic* environment. If the environment changes, the ego becomes progressively maladaptive.

Drift layer. Environmental change can occur at multiple levels: parameter drift ($\lambda_\alpha(t)$, $\gamma_\alpha(t)$), spectral-density deformation ($J(\omega, t)$), or full process-distribution shift ($P_t(X)$). For analytical tractability, we model drift at the *spectral-density parameter level* throughout this section. No monotone extension to arbitrary distribution shift is claimed.

7.1 The Rigidity Trap

Proposition 27 (Frame Rigidity under Drift)**.** *Let the environment undergo slow drift: the spectral density parameters change as* $\lambda_\alpha(t) = \lambda_\alpha^{(0)} + \varepsilon\, f_\alpha(t)$ *for* $\alpha \in \{z, x\}$*, with drift rate* $\varepsilon > 0$*. Assume the supplied distortion has a locally unique* C^1 *minimizer* $\mathcal{F}^*(\lambda) \in G$ *and* $\|D_\lambda \mathcal{F}^*\| \le K$ *along the path.*

If the agent's reference frame $\mathcal{F}$ *is held fixed (no recalibration), the mismatch between* $\mathcal{F}$ *and* $\mathcal{F}^*(t)$ *then obeys*

$$\delta(t) := d_G\big(\mathcal{F},\, \mathcal{F}^*(t)\big) \le K\varepsilon \int_0^t \|\dot{f}(s)\|\, ds, \tag{35}$$

where d_G is the geodesic distance in the gauge group.

Proof. The chain rule and the stated sensitivity bound give $\|\dot{\mathcal{F}}^*(t)\| \le K\varepsilon\|\dot{f}(t)\|$. Integrating path length bounds the geodesic distance and yields (35). □

7.2 Stylized Drift Model

To quantify the loss of visibility of a fixed frame, we use the linear drift supported by Proposition 27. No unstable environmental dynamics or Lyapunov exponent is assumed.

Definition 28 (Rotating optimal frame)**.** *Let the mismatch angle $\theta(t)$ between the agent's fixed frame $\mathcal{F}$ and the instantaneous optimal frame $\mathcal{F}^*(t)$ evolve as*

$$\theta(t) = \theta_0 + \varepsilon t, \tag{36}$$

where $\theta_0 \in (0, \pi/4)$ is the initial misalignment (so that the crossover time below is positive) and $\varepsilon > 0$ is the angular drift rate. This is a local model up to the first visibility crossover, not a claim about long-time motion on the compact gauge group.

The visible and hidden survival components decompose geometrically:

$$\mathcal{S}_{\text{vis}}(t) = \mathcal{S}_{\text{tot}}\,\cos^2\theta(t), \qquad \mathcal{S}_{\text{hid}}(t) = \mathcal{S}_{\text{tot}}\,\sin^2\theta(t), \tag{37}$$

where $\mathcal{S}_{\text{tot}}$ is the full survival functional (invariant under frame rotation).

7.3 The Prediction Error Divergence

Proposition 29 (Growth of the hidden component)**.** *Under the drift model* (36)–(37)*, the hidden survival component grows as*

$$\big|\mathcal{S}_{\text{hid}}(t)\big| = |\mathcal{S}_{\text{tot}}|\,\sin^2\big(\theta_0 + \varepsilon t\big). \tag{38}$$

For small angles ($\theta_0 + \varepsilon t \ll 1$): $|\mathcal{S}_{\text{hid}}| \approx |\mathcal{S}_{\text{tot}}|(\theta_0 + \varepsilon t)^2$.

Proof. Direct substitution of (36) into (37). The small-angle expansion $\sin^2\theta \approx \theta^2$ gives the quadratic form. □

7.4 Visibility Crossover and Conditional Gradient Reversal

Proposition 30 (Visibility crossover and a conditional gradient reversal)**.** *Under* (B1)–(B5) *with the drift model* (36) *and initial misalignment $\theta_0 \in (0, \pi/4)$, an agent with a fixed reference frame $\mathcal{F}$ reaches the visibility crossover*

$$t_{\text{vis}} = \frac{\pi/4 - \theta_0}{\varepsilon}, \tag{39}$$

at which the hidden and visible magnitudes are equal. For $t > t_{\text{vis}}$, the hidden magnitude is larger while $\theta(t) < \pi/2$. This geometric crossover alone implies neither gradient anti-correlation nor failure of every in-frame detector.

There nevertheless exist specified controllers for which gradients do reverse. In the two-mode model

$$\mathcal{S}_{\text{vis}}(u) = -\tfrac{1}{2}(u-1)^2, \qquad \mathcal{S}_{\text{full}}(u,\theta) = -\tfrac{1}{2}\|ue_1 - y^*(\theta)\|^2, \quad y^*(\theta) = \cos\theta\, e_1 + \sin\theta\, e_2, \tag{40}$$

the two scalar update directions are anti-correlated exactly when

$$\cos\theta(t) < u < 1.$$

Thus the reversal time is controller- and state-dependent; it need not coincide with t_{vis}.

Proof. The crossover $|\mathcal{S}_{\text{hid}}| = |\mathcal{S}_{\text{vis}}|$ occurs when $\sin^2\theta = \cos^2\theta$, i.e., $\theta(t_{\text{vis}}) = \pi/4$. Substituting (36): $\theta_0 + \varepsilon t_{\text{vis}} = \pi/4$, which gives (39). For (40), $\partial_u \mathcal{S}_{\text{vis}} = 1 - u$ and $\partial_u \mathcal{S}_{\text{full}} = \cos\theta - u$. Their product is negative exactly when u lies strictly between $\cos\theta$ and 1. Outside that interval the two gradients agree in sign or one vanishes. The calculation exhibits the claimed effect without promoting it to a universal consequence of visibility loss. □

Remark 31 (Model scope)**.** *For* $\theta_0 = 0$*, the crossover occurs at* $t_{\text{vis}} = \pi/(4\varepsilon)$*. With* $\varepsilon = 0.01\,\omega_0$*,* $t_{\text{vis}} \approx 79\,\omega_0^{-1}$*—long enough for the agent to accumulate a false sense of security, yet short on environmental timescales. Neither this number nor the two-mode sign condition proves that an agent cannot detect drift from residual statistics; Paper III studies one such detector.*

Remark 32 (Why dithering does not help)**.** *One might ask whether the agent could escape the delusion trap by randomly "probing" the background subspace* V_{bg}*—temporarily rotating its frame to sample hidden components. In the raw-coordinate model, a full probe competes with foreground processing at the component-count rate in Remark 8. This does not prove dithering fails: randomized exploration can reveal hidden directions, with a model-dependent cost–benefit trade-off. A deterministic in-frame controller may lack a signal indicating when or where to probe. As long as* $|\mathcal{S}_{\text{hid}}| < |\mathcal{S}_{\text{vis}}|$ *(pre-delusion), the in-frame performance metric* $\mathcal{S}_{\text{vis}}$ *shows no anomaly. The linear-drift visibility model below motivates monitoring residual statistics, the subject of Paper III, but does not rule out other exploration policies.*

Remark 33 (The ego as medicine and poison)**.** *In this model, coordinate selection reduces raw load but creates a blind subspace. The visibility crossover below motivates Paper III. This is a conditional engineering trade-off, not an irreversible logic chain.*

7.5 The Origin of Paper III

To respond to the visibility loss in this model, the agent may use a mechanism to monitor the quality of its own reference frame—to "observe its own observation." This requires a *second-order control loop*: a meta-controller that adjusts the gauge fixing σ in response to accumulated prediction errors.

The distribution of in-frame prediction residuals can itself change as the frame becomes stale. Paper III studies whether a specified likelihood model makes that change statistically identifiable. This paper does not prove that Fisher information is the unique detector or that every controller must add such a loop.

This is the subject of Paper III: the Fisher information geometry of self-referential calibration, and the thermodynamic cost of the loop that closes the chain *Chaos* → *Time* → *Self* → *Calibration*.

8 Numerical Demonstration

The preceding sections establish analytic bounds and a worked example with a qubit in a two-channel bath. We now compare budgeted coordinate-selection rules in a minimal multi-dimensional system. The observed attention concentration and selection advantage are properties of this supplied algorithm and environment. Full code and parameters are provided for reproducibility.

8.1 Model

Environment. A D-dimensional linear prediction task with sparse rotating support: $y(t) = \mathbf{w}^*(t)^\top \mathbf{x}(t) + \xi(t)$, $\mathbf{x}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$, $\xi \sim \mathcal{N}(0, \sigma^2)$. Only $m \ll D$ dimensions carry nonzero weight at any time; the active support rotates every τ_{switch} steps, modelling environmental drift.

Hard budget constraint. Per step, the agent may update only k coordinates of its weight vector (a hard processing budget), mirroring the bounded computation assumption (B2).

Agents.

- **Budgeted selector**: selects the top-k dimensions by importance score—an exponential moving average of the signed per-coordinate gradient. Signed accumulation ensures that noise dimensions (zero expected signal) cancel over time while signal dimensions persist, enabling reliable discrimination without access to the true support.
- **Random-k baseline**: selects k dimensions uniformly at random each step. This provides a budget-fair comparison with no persistent coordinate preference.

The choice of *signed* gradient EMA (rather than squared-gradient magnitude) is structurally motivated: for noise dimensions $\mathbb{E}[r\, x_i] = 0$, so the signed accumulation cancels over time; for signal dimensions $\mathbb{E}[r\, x_i] \neq 0$, so a consistent directional bias persists. The signed EMA thus acts as a *directional coherence filter* that discriminates signal from noise without access to the true support—a minimal model of a persistent coordinate preference.

Parameters.

Quantity	Value	Role
D	64	ambient dimension
m	8	signal dimensions (sparse support)
T	10,000	horizon per trial
Seeds	10	independent replications
σ	0.3	observation noise std
η	0.02	SGD learning rate
λ	0.995	weight decay per step
k	2, 4, 6, 8, 10, 12, 16, 20, 24, 32, 48, 64	budget grid
τ_{switch}	$\{500, 1000, 2000\}$	support rotation period

Attention entropy. Let n_i be the number of updates coordinate i receives in a measurement window of the last 1,000 steps. The normalised update frequency $p_i = n_i / \sum_j n_j$ defines the attention entropy:

$$H_{\text{attn}} = -\sum_{i=1}^{D} p_i \ln p_i. \tag{41}$$

Under uniform coordinate selection, $p_i = 1/D$ and $H_{\text{attn}} = \ln D$. Under budget-constrained selection, H_{attn} collapses away from $\ln D$, measuring concentration of the update rule. It is not by itself a thermodynamic order parameter.

Oracle metric. Neither agent has access to $\mathbf{w}^*(t)$. Performance is evaluated externally using the weight-space mean-squared error $\text{MSE} = \|\hat{\mathbf{w}} - \mathbf{w}^*\|^2$, averaged over post-burn-in steps.

8.2 Results

Figure 2 shows the two key signatures.

Result 1: Attention entropy collapse (Figure 2a). Under fixed support (no rotation), the attention entropy H_{attn} exhibits a sharp collapse away from $\ln D = \ln 64 \approx 4.16$ and increases monotonically with k, consistent with a budget-induced concentration of update mass onto signal-carrying dimensions. For budgets near and below the signal scale ($k \leq m$), H_{attn} remains $O(\ln m)$, consistent with confinement to the signal subspace. We use the collapse of H_{attn} away from $\ln D$ as the order parameter of symmetry breaking; a strict plateau at $\ln m$ is not expected under the present re-selection dynamics and finite-window estimator.

Result 2: Selection advantage (Figure 2b). Under rotating support, the mean-squared error gap $\Delta\text{MSE} = \text{MSE}_{\text{rnd}} - \text{MSE}_{\text{sel}}$ is positive for $k \lesssim 3m$ and peaks at tight budgets ($k = 2$) where the selection advantage is strongest. For $k \gg m$ the gap turns slightly negative (the selector's commitment to stale dimensions costs more than the random baseline's diversification), before returning to zero at $k = D$. The three τ_{switch} curves are ordered: slower drift (larger τ) yields a larger peak gap, with the ordering most visible at small k.

8.3 Scope of This Demonstration

These simulations compare two supplied coordinate-selection rules in the stated sparse linear model. They do not establish a universal symmetry-breaking transition or identify the capacity $k_{\max}$ with an information-bottleneck optimum.

This demonstration **does** show:

1. Under hard budget constraints, attention entropy collapses sharply away from $\ln D$ and remains $O(\ln m)$ for $k \leq m$—the agent confines its updates to the signal subspace. This measures concentration of the chosen update rule.

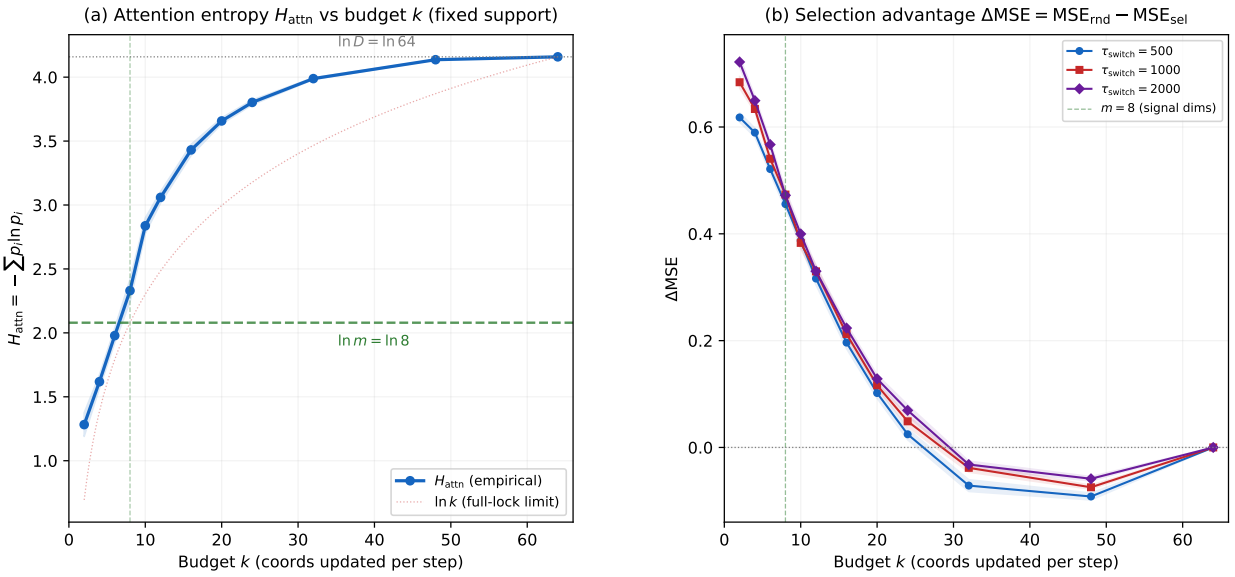


Figure 2: **Budgeted coordinate selection in the declared linear model.** $D = 64$, $m = 8$, $T = 10{,}000$, 10 seeds, 95% CI bands. **(a)** Attention entropy H_{attn} vs budget k (fixed support). The empirical curve (blue) collapses from $\ln D \approx 4.16$ toward an $O(\ln m)$ floor as budget tightens. For $k \leq m$, H_{attn} remains below $\ln m \approx 2.08$ (green dashed), consistent with confinement to the signal subspace. **(b)** Selection advantage $\Delta\text{MSE} = \text{MSE}_{\text{rnd}} - \text{MSE}_{\text{sel}}$ vs budget k under rotating support. The gap is positive for $k \lesssim 3m$ (selection helps), turns slightly negative at large k (commitment cost exceeds diversification), and returns to zero at $k = D$. Slower drift ($\tau = 2000$) yields a larger peak advantage.

2. A budgeted selector that exploits importance-weighted selection systematically outperforms a budget-fair random baseline, consistent with a survival advantage for the parameter range shown.

3. The advantage scales with both budget tightness (smaller k) and environmental stability (larger τ_{switch}).

In summary, this demonstration establishes a model-specific selection advantage and a measurable attention-concentration statistic; it does not derive selection from the budget or claim universality across architectures or environment classes.

This demonstration does **not** show:

1. That H_{attn} reaches a strict plateau at $\ln m$ for all $k \leq m$. Under the adaptive reselection dynamics used here, the selector cycles within the signal subspace, producing H_{attn} values near but not locked to $\ln m$. The relevant signature is the collapse *away from* $\ln D$, not convergence to a specific lower bound.

2. That the specific form of the importance score (signed gradient EMA) is optimal. It is one realisation of the selection mechanism.

3. That the results generalise to all environment classes. The model uses Gaussian features, linear regression, and sparse rotating support.

4. That the delusion–correction cycle is addressed. This is the subject of Paper III.

5. That a sharp thermodynamic phase transition occurs, or that $k = k_{\max}$ is the distortion-minimizing retained dimension.

Reproducibility. The complete simulation is a self-contained Python script (`paper2_kstar_scaling_demo.py`, $\sim$540 lines, requiring only NumPy and Matplotlib) with fixed random seeds. All figures in this section can be reproduced by executing the script. The following files are included in the supplementary archive:

- `paper2_kstar_scaling_demo.py` — simulation script
- `fig_paper2_kstar_scaling.pdf` — Figure 2
- `kstar_scaling_data.csv` — raw performance gap data

9 Discussion

9.1 Summary of Results

Result	**Statement**	**Sec.**
Raw-channel load	A declared D-channel archive costs Dh_μ per incremental update; this is accounting, not a lower bound	3
Compression existence	A minimizer exists in the supplied compact encoder class and has $k_{\text{opt}} \leq k_{\max}$	4.2
Conditional frame selection	A chosen proper projection breaks the declared model symmetry when no invariant feasible encoder meets the target	4.3
Four diagnostics	Frame-dependent models can be audited using $\mathcal{B}_{\text{select}}$, $\mathcal{B}_{\text{frame}}$, $\mathcal{B}_{\text{center}}$, $\mathcal{B}_{\text{inc}}$	4.4
Survival Decomposition	$\mathcal{S} = \mathcal{S}_{\text{vis}} + \mathcal{S}_{\text{hid}}$	5.2
Uniform-channel trade-off	Predictive loss is approximately $1 - k^*/D$ only under the uniform raw-channel assumption	5.4
Visibility crossover	Linear drift gives $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$; gradient reversal requires a separate controller condition	7.4
Numerical demo	Attention concentration and model-specific selection advantage (Fig. 2)	8

9.2 Testable Predictions

The conditional model claims can be tested without treating them as universal laws. First, vary k, D, and the support-switching time in the declared sparse linear environment and compare the supplied selector with random and alternative compressed baselines. A sharp transition is not predicted without a separately defined thermodynamic limit.

Second, estimate $I(X_{\text{past}}; X_{\text{future}})$ and $I(\hat{X}_{\text{past}}; X_{\text{future}})$ while varying k/D. Linear $1-k/D$ scaling tests the uniform-allocation assumption in Theorem 24; systematic curvature falsifies that assumption rather than information theory.

Third, in a controlled rotating-feature task, measure both the visibility ratio and the inner product of surrogate and oracle gradients. The linear model predicts the visibility crossover at $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$. Proposition 30 predicts gradient reversal only for controllers entering the region $\cos\theta(t) < u(t) < 1$; coincidence of the two events is not required.

9.3 What This Paper Does and Does Not Show

This paper **does** show:

1. A declared raw D-channel archive has the stated component-count load, and a minimizer exists in the supplied compact encoder class.

2. A proper coordinate projection privileges a frame; under additional feasibility assumptions this can be interpreted as operational symmetry breaking (Remark 18).

3. Definition 19 supplies four model diagnostics, with the extra controller and task assumptions needed by each made explicit.

4. Linear frame drift produces a visibility crossover, while the two-mode model exhibits a separate, conditional gradient reversal (Proposition 30).

5. A minimal computational demonstration measures attention concentration and a selection advantage over a budget-fair random baseline (Section 8, Figure 2).

This paper does **not** show:

1. That bounded computation uniquely determines a privileged basis, forces symmetry breaking, or makes the chosen frame survival-optimal.

2. That symmetry breaking is necessary or sufficient for persistence.

3. That the "ego" implies or requires consciousness, subjective experience, or phenomenal awareness. The term is used strictly in the control-theoretic sense.

4. That this framework constitutes a theory of consciousness. It is a theory of computational optimality under thermodynamic constraints.

5. That the four diagnostics exhaust self-reference or follow from gauge fixing alone.

6. That the coordinate capacity is a Shannon rate–distortion bound or is achievable by a specific physical implementation.

7. That visibility loss forces gradient reversal or is undetectable. Paper III analyzes one conditional calibration model.

8. That the framework constitutes or implies a philosophical or metaphysical claim about the nature of selfhood.

9. That this framework applies to all possible physical systems. It applies to systems satisfying the declared channel, encoder, controller, and drift assumptions.

10. That the Clifford algebra is the only possible algebraic setting. It is supplied here as a model input; Q-RAIF does not uniquely derive it.

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